LEARNING SESSION S3

Power BI: Basic to Advanced

From your first report to DAX, data modelling, Copilot, Fabric and deployment, updated for 2026

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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📈 Power BI: Basic to Advanced

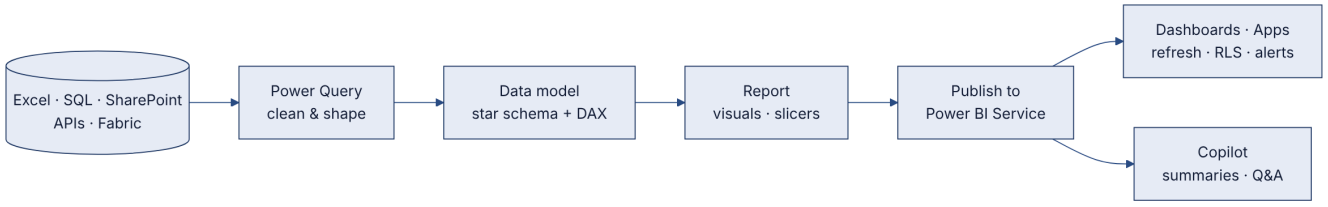
From your first report to DAX, data modelling, Copilot, Fabric and deployment, updated for 2026

Learning session · non-tech friendly · aligned to the PL-300 Power BI Data Analyst certification

Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#s3**

Roadmap

Power BI turns spreadsheets and databases into interactive dashboards that refresh themselves. This session builds the skill in seven steps; the course notes in **Power BI (course module)** stay the quick reference.



The Power BI flow: get data → Power Query → model + DAX → report → publish → share, refresh, Copilot.

Step	Skill	Milestone
1	Install, interface, first report	A 1-page sales report in 30 minutes
2	Power Query	Clean and combine monthly files automatically
3	Data modelling	Star schema with a date table
4	DAX: basic → advanced	Time intelligence, ranking, dynamic measures
5	Visual design & interactivity	Drill-through, bookmarks, field parameters
6	Service, Fabric & security	Published app with scheduled refresh and RLS
7	Copilot, performance, DevOps	Fast, AI-assisted, version-controlled reports

1 · First report

1. Install **Power BI Desktop** free from the Microsoft Store (Windows). Mac users can use the Power BI service in a browser or a Windows VM.
2. **Get data → Excel workbook** → tick the Sales table → **Transform Data** (always check types first) → Close & Apply.
3. Report view: click a **Card** visual, drag Amount → shows Total Sales.
4. Add a **Clustered column chart**: Axis = City, Values = Amount; sort descending.

5. Add a **Line chart**: X-axis = Order Date (Month), Values = Amount.
6. Add a **Slicer** on Store Type; click values and watch all visuals filter together (cross-filtering).
7. Format: theme (View → Themes), titles that state the insight, consistent colours.
8. Save as .pbix; Publish to a workspace to share.

Pane / view	Purpose
Report view	Canvas and pages
Table view	Look at rows, set formats, add calculated columns
Model view	Relationships and properties
DAX query view / TMDL view	Write and test DAX queries; edit the model as code
Visualizations / Data / Filters panes	Choose visuals, fields, filter levels

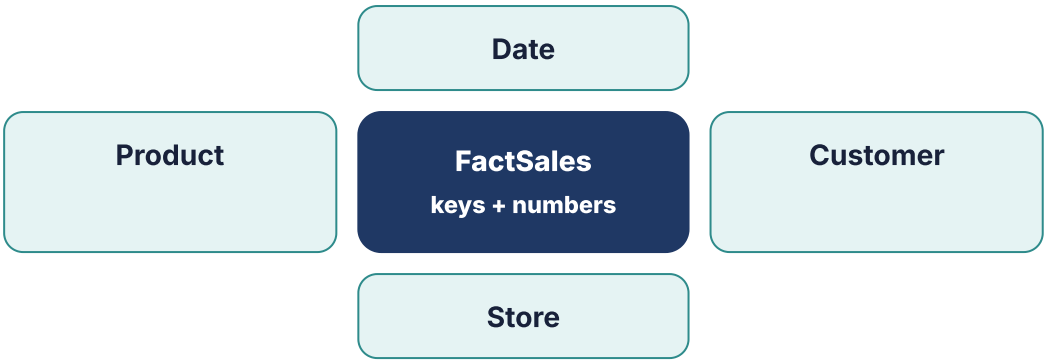
2 · Power Query

Home → Transform data. Every action becomes an **Applied Step**; refreshing replays them on new data.

Need	Do this
Combine 12 monthly files	Get Data → Folder → Combine & Transform
Fix types	Icon left of each header; set Date, Decimal, Text (locale for dd-mm-yyyy: Using Locale → English (India))
Wide months to rows	Select ID columns → Unpivot Other Columns
Lookup another table	Merge Queries (Left Outer) → expand the needed column
Stack tables	Append Queries
Fill blanks from above	Transform → Fill → Down
Categories without code	Add Column → Conditional Column
Data quality check	View → Column quality, distribution, profile (based on entire data)
Parameterise a source path	Manage Parameters → use in Source step

> **Tip** **Query folding**: with databases, Power Query translates steps into SQL that runs on the server. Keep foldable steps (filters, column selection) early; right-click a step → View Native Query to check.

3 · Data modelling



Star schema: one fact table in the middle, dimension tables around it, one-to-many relationships flowing from dimension to fact.

- Facts = events with numbers (sales lines, transactions). Dimensions = descriptions (who, what, where, when).
- Relationships: one-to-many, single direction by default; avoid bi-directional unless needed.
- **Date table**: one row per day covering the full range; mark as date table; use it for all time slicing.
- Hide technical columns (keys) from report view; set sort-by columns (Month name sorted by month number); set default summarisation; organise measures in a measure table.
- Role-playing dates (order vs ship date): one active relationship, others activated with `USERELATIONSHIP`.

4 · DAX

Level 1: basic measures

```
Total Sales   = SUM ( Sales[Amount] )
Orders        = DISTINCTCOUNT ( Sales[OrderID] )
AOV           = DIVIDE ( [Total Sales], [Orders] )
Total Cost    = SUMX ( Sales, Sales[Qty] * RELATED ( Product[UnitCost] ) )
Margin %      = DIVIDE ( [Total Sales] - [Total Cost], [Total Sales] )
```

Level 2: CALCULATE and filter context

Every visual cell evaluates a measure under filters from rows, columns, slicers and filters. **CALCULATE** changes those filters.

```
Sales Pune        = CALCULATE ( [Total Sales], Customer[City] = "Pune" )
Sales All Cities  = CALCULATE ( [Total Sales], ALL ( Customer[City] ) )
% of All Cities   = DIVIDE ( [Total Sales], [Sales All Cities] )
Sales Ignoring Slicers Except Year = CALCULATE ( [Total Sales], ALLEXCEPT ( 'Date', 'Date'[Year]
) )
Big Orders        = CALCULATE ( [Orders], Sales[Amount] > 10000 )
```

Level 3: time intelligence

```
Sales YTD      = TOTALYTD ( [Total Sales], 'Date'[Date], "31/3" )          -- Indian FY ends 31
March
Sales LY       = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( 'Date'[Date] ) )
YoY %          = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )
Sales Last 3M  = CALCULATE ( [Total Sales], DATESINPERIOD ( 'Date'[Date], MAX ( 'Date'[Date] ),
-3, MONTH ) )
MoM %          = VAR prev = CALCULATE ( [Total Sales], DATEADD ( 'Date'[Date], -1, MONTH ) )
                 RETURN DIVIDE ( [Total Sales] - prev, prev )
```

Level 4: advanced patterns

```
Product Rank   = RANKX ( ALLSELECTED ( Product[Name] ), [Total Sales], , DESC, DENSE )
Top 5 Sales    = CALCULATE ( [Total Sales], TOPN ( 5, ALL ( Product[Name] ), [Total Sales] ) )
New Customers  = COUNTROWS ( FILTER ( VALUES ( Sales[CustomerID] ),
                     CALCULATE ( MIN ( Sales[Date] ) ) >= MIN ( 'Date'[Date] ) ) )
Pareto Cum %   = VAR cur = [Total Sales]
                 VAR tbl = ADDCOLUMNS ( ALLSELECTED ( Product[Name] ), "s", [Total Sales] )
                 RETURN DIVIDE ( SUMX ( FILTER ( tbl, [s] >= cur ), [s] ), SUMX ( tbl, [s] ) )
Selected Metric = SWITCH ( SELECTEDVALUE ( Metric[Name] ),
                     "Sales", [Total Sales], "Margin %", [Margin %], [Orders] )
```

Visual calculations (generally available since May 2026)

Select a visual → **New visual calculation**. Write simpler DAX that runs on the visual's own grid, without touching the model:

```
Running Total = RUNNINGSUM ( [Sales] )
Moving Avg 3  = MOVINGAVERAGE ( [Sales], 3 )
% of Parent   = DIVIDE ( [Sales], COLLAPSE ( [Sales], ROWS ) )
vs Previous   = [Sales] - PREVIOUS ( [Sales] )
```

> **Tip** Rule of thumb: model measures for numbers reused across reports; visual calculations for one-off running totals, ranks and comparisons in a single visual.

5 · Visuals & interactivity

Cards / KPI / Gauge	Headline numbers vs target
Bar, column, line, combo, area, ribbon, waterfall, funnel, scatter, map	Standard analysis visuals
Matrix	Pivot-style table with drill-down, conditional formatting and custom totals
Decomposition tree, Key influencers, Smart narrative	AI visuals that explain a metric

Small multiples	One chart repeated per category
Gantt (preview, Oct 2026)	Project timelines natively

Interactivity

- **Edit interactions**: filter vs highlight vs none per visual pair.
- **Drill-through** pages and **tooltip pages**.
- **Bookmarks** + Selection pane + Buttons: toggles, guided stories, reset filters.
- **Field parameters**: let users switch the measure or the axis.
- **Slicers**: list, dropdown, button slicer, date range; sync across pages.
- **Personal bookmarks** and personalise visuals for viewers in the service.

Design checklist

✔ Top-left = most important KPI; Z-pattern layout
✔ Max 6–8 visuals per page; one message per page
✔ Colour has meaning (red = bad), consistent theme JSON
✔ Accessible: contrast, alt text, tab order
✔ Titles state insights; units in labels (₹ lakh)

6 · Service, Fabric & security

- **Workspaces** (team folders) with roles Admin / Member / Contributor / Viewer; **Apps** package reports for wide audiences.
- **Semantic model** (dataset) settings: credentials, scheduled refresh (up to 8/day on Pro, 48 on Premium/Fabric capacity), **incremental refresh** for large facts, on-premises data gateway.
- **Row-level security**: roles with DAX filters such as `[Region] = "North"` or dynamic `[Email] = USERPRINCIPALNAME()`; assign users in the service. Object-level security hides tables/columns.
- **Dashboards** with pinned tiles and data alerts; **subscriptions** by email; export to PDF/PowerPoint; Analyze in Excel.
- **Microsoft Fabric**: the wider platform (OneLake, Lakehouse, Warehouse, Data Factory pipelines, notebooks, Direct Lake mode). Microsoft is moving enterprise capacity from Premium P-SKUs to Fabric F-SKUs.
- **Licences**: Free (personal use), Pro (share with Pro users), Premium Per User, Fabric capacity (share with free viewers on F64 and above).

7 · Copilot, performance & DevOps

Copilot in Power BI

Report creation	"Create a page showing sales trend, top products and regional split"
Summaries	Copilot Summarize at report and visual level in the service (May 2026)

Q&A over the model	Ask questions in English; answers respect RLS
DAX help	Write and explain measures in the DAX query view
Prep data for AI	Add synonyms, descriptions and verified answers so Copilot answers correctly

Copilot needs a paid Fabric capacity (F2 or above) or Premium, and an admin must enable it.

Performance

- View → **Performance Analyzer**: time per visual (DAX query, visual display).
- Smaller model = faster: remove unused columns, reduce cardinality (split datetime into date and time), prefer measures over calculated columns, integers for keys.
- External tools: DAX Studio (query timing, VertiPaq Analyzer), Tabular Editor (bulk editing, best-practice analyser).

Version control and deployment

- Save as **Power BI Project (.pbip)**: report (PBIR format) and model (**TMDL**) as text files that work with Git.
- Deployment pipelines: Development → Test → Production workspaces.
- Fabric Git integration with Azure DevOps or GitHub.

Interview Q&A

Q Measure vs calculated column?

A column is computed per row at refresh and stored (use for slicing); a measure is computed at query time in the current filter context (use for aggregation).

Q Explain filter context vs row context.

Filter context = filters applied to the model by the visual, slicers and CALCULATE. Row context = the current row inside a calculated column or an iterator (SUMX). CALCULATE converts row context into filter context (context transition).

Q What is a star schema and why use it?

A central fact table linked one-to-many to dimension tables. It keeps DAX simple and fast and avoids ambiguous filter paths.

Q Import vs DirectQuery vs Direct Lake?

Import: data cached in memory, fastest, needs refresh. DirectQuery: live queries to the source, current but slower. Direct Lake (Fabric): reads Delta tables in OneLake directly with import-like speed.

Q How do you implement dynamic RLS?

Have a user–region mapping table, relate it to the model, and use a role filter like `[Email] = USERPRINCIPALNAME()`.

Q What are visual calculations?

DAX calculations defined on a visual that operate on the visual's data grid (running sums, moving averages, previous value) without adding model measures.

Q How would you speed up a slow report?

Use Performance Analyzer to find slow visuals; optimise DAX (variables, avoid FILTER on whole tables), reduce model size, limit visuals per page, use aggregations or incremental refresh.

Resources

- ▶ YouTube: Simplilearn Power BI full course · ▶ YouTube: Simplilearn Power BI tutorial for beginners
- ▶ YouTube: Guy in a Cube Power BI · ▶ YouTube: SQLBI CALCULATE explained · ▶ YouTube: Power BI visual calculations tutorial
- ▶ YouTube: Pragmatic Works Power BI full course · ▶ YouTube: codebasics Power BI project
- 📘 Microsoft Learn: Power BI · 📘 PL-300 study guide
- 📘 Power BI monthly feature summaries (blog) · 📘 DAX Guide